

Bacteroidetes/Firmicutes ratio

Muluh

Bacteroidetes/Firmicutes ratio

Group-wise comparisons

- Taxonomic level: Phylum
- Group variable: meal_group
- Multiple testing correction: fdr
- Statistical test Wilcoxon

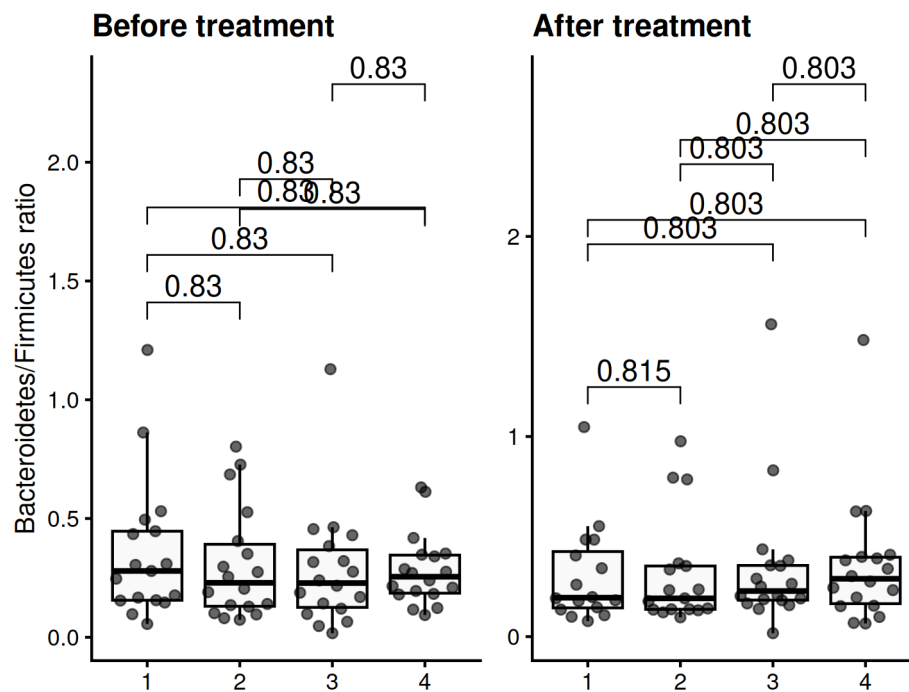


Table of the top findings

Across-group

Before treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of Bacteroidetes/Firmicutes before Treatment

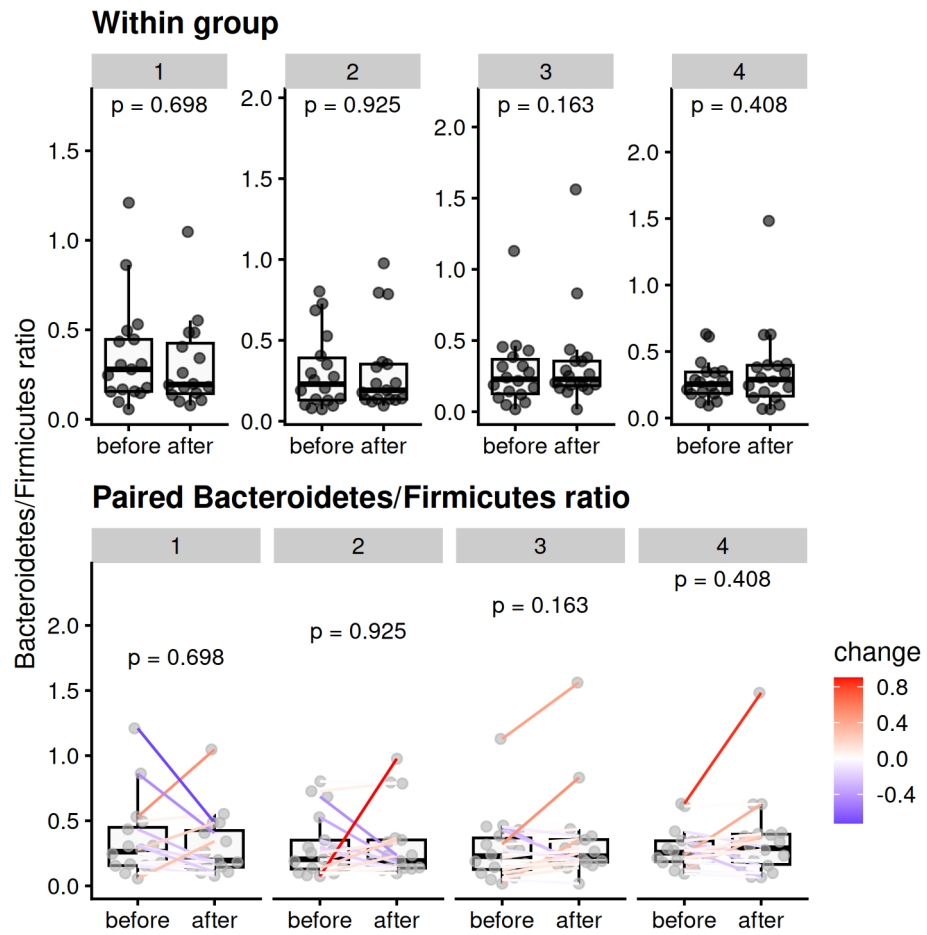
group1	group2	mean1	mean2	log2FC	p.adj
1	2	0.36	0.30	-0.23	0.83
1	3	0.36	0.28	-0.34	0.83
1	4	0.36	0.28	-0.34	0.83
2	3	0.30	0.28	-0.11	0.83
2	4	0.30	0.28	-0.10	0.83
3	4	0.28	0.28	0.00	0.83

After treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 2: Table of Bacteroidetes/Firmicutes after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
1	2	0.31	0.31	0.04	0.815
1	3	0.31	0.34	0.17	0.803
1	4	0.31	0.36	0.23	0.803
2	3	0.31	0.34	0.12	0.803
2	4	0.31	0.36	0.19	0.803
3	4	0.34	0.36	0.06	0.803



Within-group

Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Table 3: Table of Bacteroidetes/Firmicutes paired between timepoint (1 and 2)

meal_group	mean1	mean2	log2FC	p.value
1	0.36	0.31	0.27	0.698
2	0.30	0.31	0.91	0.925
3	0.28	0.34	1.21	0.163
4	0.28	0.36	0.37	0.408

Data Summary

```

class: TreeSummarizedExperiment
dim: 5 140
metadata(1): agglomerated_by_rank
assays(2): counts relabundance
rownames(5): Actinobacteria Bacteroidetes Firmicutes Other
             Proteobacteria
rowData names(10): kingdom phylum ... clade_name phylum_sub
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(14): sample id ... group bact_firm_ratio
reducedDimNames(0):
mainExpName: NULL
altExpNames(0):
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL

```

Version info

```

R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.1 LTS

```

```

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0

```

```

locale:

```

```

[1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
[3] LC_TIME=en_US.UTF-8       LC_COLLATE=en_US.UTF-8
[5] LC_MONETARY=en_US.UTF-8   LC_MESSAGES=en_US.UTF-8
[7] LC_PAPER=en_US.UTF-8      LC_NAME=C
[9] LC_ADDRESS=C              LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

```

```

time zone: Europe/Helsinki
tzcode source: system (glibc)

```

attached base packages:

```

[1] stats4      grid      stats      graphics  grDevices  utils      datasets
[8] methods     base

```

other attached packages:

```

[1] kableExtra_1.4.0          vegan_2.6-8
[3] lattice_0.22-6            permute_0.9-7
[5] lubridate_1.9.3           forcats_1.0.0
[7] stringr_1.5.1             purrr_1.0.2
[9] readr_2.1.5              tibble_3.2.1
[11] tidyverse_2.0.0          tidyr_1.3.1
[13] sechm_1.13.1             ComplexHeatmap_2.21.1
[15] scatter_1.33.4           scuttle_1.15.4
[17] quarto_1.4.4             patchwork_1.3.0
[19] parameters_0.24.0        multtest_2.61.0
[21] miaTime_0.1.21           miaViz_1.15.4
[23] ggraph_2.2.1             mia_1.15.6
[25] TreeSummarizedExperiment_2.13.0 Biostrings_2.73.2
[27] XVector_0.45.0           SingleCellExperiment_1.27.2
[29] MultiAssayExperiment_1.31.5 SummarizedExperiment_1.35.3
[31] Biobase_2.65.1           GenomicRanges_1.57.1
[33] GenomeInfoDb_1.41.2      IRanges_2.39.2
[35] S4Vectors_0.43.2         BiocGenerics_0.51.3
[37] MatrixGenerics_1.17.0    matrixStats_1.4.1
[39] gridExtra_2.3            gtsummary_2.0.4
[41] ggsignif_0.6.4           ggpubr_0.6.0
[43] ggplot2_3.5.1           DT_0.33
[45] dplyr_1.1.4             cardx_0.2.2
[47] ANCOMBC_2.8.0

```

loaded via a namespace (and not attached):

```

[1] randomcoloR_1.1.0.1      gld_2.6.6
[3] nnet_7.3-19              TH.data_1.1-2

```

[5] vctrs_0.6.5	energy_1.7-12
[7] digest_0.6.37	png_0.1-8
[9] shape_1.4.6.1	proxy_0.4-27
[11] Exact_3.3	registry_0.5-1
[13] rbiom_1.0.3	ggrepel_0.9.6
[15] bayestestR_0.15.0	mediation_4.5.0
[17] MASS_7.3-61	reshape2_1.4.4
[19] foreach_1.5.2	withr_3.0.2
[21] xfun_0.49	ggfun_0.1.6
[23] survival_3.7-0	doRNG_1.8.6
[25] memoise_2.0.1	ggbeeswarm_0.7.2
[27] emmeans_1.10.4	systemfonts_1.1.0
[29] gmp_0.7-5	tidytree_0.4.6
[31] zoo_1.8-12	GlobalOptions_0.1.2
[33] gtools_3.9.5	V8_5.0.1
[35] Formula_1.2-5	datawizard_0.13.0
[37] httr_1.4.7	rstatix_0.7.2
[39] ps_1.8.0	rstudioapi_0.17.1
[41] UCSC.utils_1.1.0	generics_0.1.3
[43] base64enc_0.1-3	processx_3.8.4
[45] curl_5.2.3	zlibbioc_1.51.1
[47] ScaledMatrix_1.13.0	polyclip_1.10-7
[49] ca_0.71.1	GenomeInfoDbData_1.2.13
[51] SparseArray_1.5.41	xtable_1.8-4
[53] doParallel_1.0.17	evaluate_1.0.1
[55] S4Arrays_1.5.10	hms_1.1.3
[57] irlba_2.3.5.1	colorspace_2.1-1
[59] readxl_1.4.3	magrittr_2.0.3
[61] later_1.3.2	viridis_0.6.5
[63] ggtree_3.13.1	DECIPHER_3.1.4
[65] cowplot_1.1.3	class_7.3-22
[67] Hmisc_5.2-0	pillar_1.9.0
[69] nlme_3.1-166	iterators_1.0.14
[71] decontam_1.25.0	compiler_4.4.2
[73] beachmat_2.21.6	stringi_1.8.4
[75] DescTools_0.99.57	TSP_1.2-4
[77] tokenizers_0.3.0	minqa_1.2.8
[79] plyr_1.8.9	crayon_1.5.3
[81] abind_1.4-8	gridGraphics_0.5-1
[83] graphlayouts_1.2.0	bit_4.5.0
[85] rootSolve_1.8.2.4	sandwich_3.1-1
[87] codetools_0.2-20	multcomp_1.4-26
[89] BiocSingular_1.21.4	slam_0.1-54

[91] e1071_1.7-16	lmom_3.2
[93] GetoptLong_1.0.5	tidytext_0.4.2
[95] splines_4.4.2	circlize_0.4.16
[97] Rcpp_1.0.13-1	sparseMatrixStats_1.17.2
[99] cellranger_1.1.0	knitr_1.48
[101] utf8_1.2.4	clue_0.3-65
[103] lme4_1.1-35.5	fs_1.6.5
[105] checkmate_2.3.2	DelayedMatrixStats_1.27.3
[107] Rdpack_2.6.1	expm_1.0-0
[109] gsl_2.1-8	ggplotify_0.1.2
[111] estimability_1.5.1	Matrix_1.7-0
[113] svglite_2.1.3	tzdb_0.4.0
[115] lpSolve_5.6.21	tweenr_2.0.3
[117] pkgconfig_2.0.3	tools_4.4.2
[119] cachem_1.1.0	rbibutils_2.2.16
[121] viridisLite_0.4.2	DBI_1.2.3
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[129] coda_0.19-4.1	insight_1.0.0
[131] carData_3.0-5	rpart_4.1.23
[133] farver_2.1.2	tidygraph_1.3.1
[135] mgcv_1.9-1	yaml_2.3.10
[137] foreign_0.8-87	cli_3.6.3
[139] lifecycle_1.0.4	mvtnorm_1.3-2
[141] bluster_1.15.1	backports_1.5.0
[143] BiocParallel_1.39.0	timechange_0.3.0
[145] gtable_0.3.6	rjson_0.2.23
[147] parallel_4.4.2	ape_5.8
[149] SnowballC_0.7.1	CVXR_1.0-15
[151] jsonlite_1.8.9	seriation_1.5.6
[153] bit64_4.5.2	Rtsne_0.17
[155] yulab.utils_0.1.8	BiocNeighbors_1.99.1
[157] RcppParallel_5.1.9	janeaustenr_1.0.0
[159] lazyeval_0.2.2	htmltools_0.5.8.1
[161] tinytex_0.54	glue_1.8.0
[163] treeio_1.29.1	boot_1.3-31
[165] igraph_2.1.1	R6_2.5.1
[167] labeling_0.4.3	Rmpfr_1.0-0
[169] cluster_2.1.8	rngtools_1.5.2
[171] aplot_0.2.3	nloptr_2.1.1
[173] DirichletMultinomial_1.47.0	DelayedArray_0.31.13
[175] tidyselect_1.2.1	vipor_0.4.7

[177]	htmlTable_2.4.3	xml2_1.3.6
[179]	ggforce_0.4.2	car_3.1-3
[181]	rsvd_1.0.5	munsell_0.5.1
[183]	data.table_1.16.2	htmlwidgets_1.6.4
[185]	RColorBrewer_1.1-3	rlang_1.1.4
[187]	lmerTest_3.1-3	ggnewscale_0.5.0
[189]	fansi_1.0.6	beeswarm_0.4.0